

Mohammad Mahmudul Hasan

AI and photogrammetry | Team Lead, AI Engineer

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RESEARCH PROFILE

AI engineering team lead with hands-on photogrammetry and mapping experience spanning AT-related workflows, DSM/DTM preparation, orthorectification, seamlines, tiling, bridge editing and final mosaics, alongside point-cloud alignment, segmentation and classification. Seeking research at the intersection of photogrammetry, 3D perception and AI.

EDUCATION

Islamic University of Technology, Bangladesh | BSc in Electrical and Electronic Engineering, completed 22 March 2021.

CGPA 3.76/4.00; first class with honours. Project and Thesis: A+; Digital Signal Processing: A; Artificial Neural Networks and Fuzzy Logic: A+.

PROFESSIONAL EXPERIENCE

HawarIT Limited — Team Lead, AI Engineer | with the company since 2021; team-lead role since [ADD month and year of promotion].

Currently lead a six-member AI engineering team.

- Worked across AT-related processing, DSM/DTM workflows, image rectification, seamline preparation, tiling, bridge editing and final single-file orthophoto output.
- Worked on point-cloud processing, alignment across multiple trajectories, segmentation and classification.
- Connected produced imagery with downstream GIS digitization and building-change analysis; currently lead a six-member AI engineering team.

Earlier titles, exact dates and individual project ownership: [ADD verified career chronology].

PEER-REVIEWED PUBLICATION

Tasnim Sakib Apon, Mohammad Mahmudul Hasan, Abrar Islam, and Md. Golam Rabiul Alam. “Demystifying Deep Learning Models for Retinal OCT Disease Classification using Explainable AI.” 2021 IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE), 2021. DOI: [10.1109/CSDE53843.2021.9718400](https://doi.org/10.1109/CSDE53843.2021.9718400).

Individual contribution: [ADD precisely what you did]. The author preprint is the same work.

SELECTED RESEARCH AND TECHNICAL WORK

FROM AERIAL IMAGERY TO ORTHOPHOTO DELIVERY | Professional experience

Worked across AT-related processing, DSM and DTM workflows, image rectification, seamline preparation, image tiling, bridge editing and final single-file output.

Research question: Which intermediate quality signals best predict visible mosaic defects or errors in downstream building extraction?

POINT-CLOUD ALIGNMENT AND SEMANTIC PROCESSING | Professional experience

Worked on point-cloud processing, alignment across multiple trajectories, segmentation and classification.

Research question: Can semantic consistency help detect alignment failures that are not obvious from aggregate geometric residuals?

BUILDING CHANGE ACROSS ACQUISITION YEARS | Professional experience

Worked with multi-year orthophotos to identify building construction, demolition and extension.

Research question: How can geometric checks and temporal reasoning reduce false change detections across surveys?

RESEARCH DIRECTION

How should uncertainty in alignment and reconstruction influence downstream object extraction and map updates?

Evaluate alignment quality together with semantic consistency, then study whether quality-aware selection or weighting reduces mapping errors. Use independent checkpoints and repeat-survey evaluation where suitable reference data are available.

TECHNICAL FOUNDATION

Orthophoto production; AT-related workflows; DSM/DTM; orthorectification; seamlines and mosaicking; point-cloud alignment, segmentation and classification; Python; MATLAB. Photogrammetry software and registration methods: [ADD verified tools].

ADDITIONAL CONTEXT

Independent AI/ML study during the final semesters of the EEE degree and participation in AI competitions. Historical Coursera credentials and competition placement should be added with year and verification links.

Language: Bangla (native); English-medium instruction certificate available. IELTS pending; intended overall score 7.0 is not an achieved score.

References: [ADD academic and professional referees as required by the program].